BASE CHANGE BY A UNIVERSAL HOMEOMORPHISM IDENTIFIES RELATIVE CYCLES

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file relative-cycles.tex; no statement or proof step was rewritten, reordered or removed. The counted span is the author environment and its own proof environment. References inside the span are handed to the assembler as navigation stubs that carry the canonical tag of the cited tag only; whether each of them is load-bearing is recorded on the stub itself.

Allowed standard prerequisites (editorial).

- Bijection of families of r-cycles on fibres under pullback along a universal homeomorphism bijective on residue fields (relative-cycles chapter, Lemma lemma-pullback-universally-bijective, canonical tag 0H4Y). cited at proof line 1508. Gives the bijection between the group of families of r-cycles on fibres of X/S and of X'/S'.
- Seminormalization section of the Morphisms chapter (Morphisms, Section morphisms-section-seminormalization, canonical tag 0EUK). cited at proof lines 1518-1519. Since R is a seminormal ring, the universal homeomorphism S' x_S Spec(R) to Spec(R) inducing bijections on residue fields has a section.

Source reference. Original lemma 0H4Y: sources/stacks-chow/tags.txt, source lines 21368–21368. Cited at proof line 1508: supplies the underlying bijection between families of r-cycles on the fibres of X/S and X'/S'.

Source reference. Original section 0EUK: sources/stacks-chow/tags.txt, source lines 18555–18555. Cited at proof lines 1518-1519: R seminormal gives the section of the universal homeomorphism S' x_S Spec(R) -> Spec(R).

Lemma 0.2. *Let $g : S' \to S$ be a universal homeomorphism of locally Noetherian schemes which induces isomorphisms of residue fields. Let $f : X \to S$ be locally of finite type. Set $X' = S' \times_S X$. Let $r \geq 0$. Then base change by g determines a bijection $z(X/S, r) \to z(X'/S', r)$.*

Proof. By Lemma 0H4Y we have a bijection between the group of families of r-cycles on fibres of X/S and the group of families of r-cycles on fibres of X'/S'. Say α is a families of r-cycles on fibres of X/S and $\alpha' = g^*\alpha$ is the base change. If R is a discrete valuation ring, then any morphism $h : \mathrm{Spec}(R) \to S$ factors as $g \circ h'$ for some unique morphism $h' : \mathrm{Spec}(R) \to S'$. Namely, the morphism $S' \times_S \mathrm{Spec}(R) \to \mathrm{Spec}(R)$ is a universal homomorphism inducing bijections on residue fields, and hence has a section (for example because R is a seminormal ring, see Morphisms, Section 0EUK). Thus the condition that α is compatible with

Date: stacks-project:relative-cycles.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

specializations (i.e., is a relative r-cycle) is equivalent to the condition that α' is compatible with specializations. $\square$